CPSC 4970 – C++ Programming
Homework 5

Your solution to each problem should include the C++ source code and the screenshot of a test run.

1. (Weather Statistics) Write a program that uses a structure to store the following weather data for a particular month:

Total Rainfall
High Temperature
Low Temperature
Average Temperature

The program should have an array of 12 structures to hold weather data for an entire year. When the program runs, it should ask the user to enter data for each month. (The average temperature should be calculated.) Once the data are entered for all the months, the program should calculate and display the average monthly rainfall, the total rainfall for the year, the highest and lowest temperatures for the year (and the months they occurred in), and the average of all the monthly average temperatures.

2. (Weather Statistics Modification) Modify the program of problem 1 so it defines an enumerated data type with enumerators for the months (JANUARY, FEBRUARY, so on). The program should use the enumerated type to step through the elements of the array.

3. (Course Grade) Write a program that uses a structure to store the following data:

Member Name	**Description**
name	student name
id	student ID number
tests	pointer to an array of test scores
average	average test score
grade	course grade

The program should keep a list of test scores for a group of students. It should ask the user how many test scores there are to be and how many students there are. It should then dynamically allocate an array of structures. Each structure's `tests` member should point to a dynamically allocated array that will hold the test scores.

After the arrays have been dynamically allocated, the program should ask for the name, ID number, and all the test scores for each student. The average test score should be calculated and stored in the `average` member of each structure. The course grade should be computed on the basis of the following grading scale:

Average Test Grade	Course Grade
91-100	A
81-90	B
71-80	C
61-70	D
60 or below	F

The course grade should then be stored in the `grade` member of each structure. Once all this data is calculated, a table should be displayed on the screen listing each students' name, ID number, average test score, and course grade.

4. **(Movie Class)** Design a class `Movie` that contains information about a movie. The class has the following attributes (member variables):

- The movie name
- The MPAA rating (for example, G, PG, PG-13, R)
- The number of people that have rated this movie as a 1 (Terrible)
- The number of people that have rated this movie as a 2 (Bad)
- The number of people that have rated this movie as a 3 (OK)
- The number of people that have rated this movie as a 4 (Good)
- The number of people that have rated this movie as a 5 (Great)

The class should have the following member functions:

- A constructor that allows the programmer to create the object with a specified name and MPAA rating. The number of people rating the movie should be set to 0 in this constructor.
- Accessor and mutator functions for the movie name and MPAA rating
- A function `addRating` that takes an integer as an input parameter. The function should verify that the parameter is a number between 1 and 5, and if so, increment the number of people rating the movie that matches the input parameter. For example, if 3 is the input parameter, then the number of people that rated the movie as a 3 should be incremented by 1.
- A function `getAverage` that returns the average value for all of the movie ratings

Test the class by writing a main function that creates an array of movie objects by calling the constructor, adds at least five ratings for each movie (ideally the ratings are from the user input), and outputs the movie name, MPAA rating, and average rating for each movie object.

5. **(Car Class)** Write a class named `Car` that has the following member variables:

- yearModel – an int that holds the car's year model
- make – a string that holds the make of the car
- speed – an int that holds the car's current speed

In addition, the class should have the following constructor and other member functions:

- Constructor – The constructor should accept the car's year model and make as arguments. These values should be assigned to the object's yearModel and make member variables. The constructor should also assign 0 to the speed member variable.

- Accessor – appropriate accessor functions to get the values stored in an object's yearModel, make, and speed member variables
- Mutator – appropriate mutator functions to set the values stored in an object's yearModel, make, and speed member variables
- accelerate – The accelerate function should add a specified value to the speed member variable each time it is called
- brake – The brake function should subtract a specified value from the speed member variable each time it is called. Make sure that the speed of the car never goes under zero.

Demonstrate the class in a program that creates a `Car` object, then calls the `accelerate` function five times. After each call to the `accelerate` function, get the current speed of the car and display it. Then, call the `brake` function five times. After each call to the `brake` function, get the current speed of the car and display it.

6. **(Trivia Game)** In this programming challenge, you will create a simple trivia game for two players. The program will work like this:

- Starting with player 1, each player gets a turn at answering five trivia questions. (There are a total of 10 questions.) When a question is displayed, four possible answers are also displayed. Only one of the answers is correct, and if the player selects the correct answer, he or she earns a point.
- After answers have been selected for all of the questions, the program displays the number of points earned by each player and declares the player with the highest number of points the winner.

In this program, you will design a `Question` class to hold the data for a trivia question. The `Question` class should have member variables for the following data:

- A trivia question
- Possible answer #1
- Possible answer #2
- Possible answer #3
- Possible answer #4
- The number of the correct answer (1, 2, 3, or 4)

The `Question` class should have appropriate constructor(s), accessor, and mutator functions.

The program should create an array of 10 `Question` objects, one for each trivia question. Make up your own trivia questions on the subject or subjects of your choice for the objects.